

Social Media Text as Data

Exploring the Potential and Limitations of Automated Content Analysis Methods

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Section 1

Part I: Why Social Media Text Matters

The Data Revolution in Marketing Research

Traditional data sources relied on:

- Surveys and focus groups (slow, expensive, self-reported)
- Sales and transaction records (behavioral, but limited)
- Panel data (structured, but retrospective)

Social media changes everything:

- **Scale:** Billions of posts daily
- **Speed:** Real-time signal
- **Naturalistic:** Unsolicited opinions
- **Richness:** Emotional, contextual
- **Accessible:** APIs, scrapers
- **Longitudinal:** Historical archives

Source: Miller (2015), Ch. 10 | Jimmy Zhenning Xu, Ph.D. | github.com/utjimmyx

Platforms and Their Data Ecosystems

Platform	Primary Format	API Access	Best For
Twitter/X	Short text	Restricted (v2)	Public opinion
Reddit	Long-form	Free (pushshift)	Niche communities
Bluesky	Short text	Open (AT Protocol)	Emerging discourse
YouTube	Comments	YouTube Data API v3	Product feedback
LinkedIn	Professional	Limited	B2B signals
TikTok	Video + captions	Restricted	Gen Z trends

Each platform requires platform-specific scraping strategy and ethical considerations.

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What Can We Learn from Social Media Text?

Key constructs researchers extract:

- 1 **Sentiment** — positive, negative, neutral valence
- 2 **Topics** — what issues are being discussed?
- 3 **Emotions** — anger, joy, fear, anticipation (NRC)
- 4 **Brand associations** — co-occurring terms and entities
- 5 **Behavioral signals** — purchase intent, complaints, advocacy
- 6 **Network structure** — who talks to whom, influence flow

Key Question

Can machines reliably measure these constructs at scale — and when do they fail?

Section 2

Part II: Core Methods of Automated Text Analysis

The Automated Content Analysis Toolkit

Conceptual overview - not executable

```
methods <- list(  
  lexicon_based = c("AFINN", "Bing", "NRC", "VADER"),  
  ml_supervised = c("Naive Bayes", "SVM", "Random Forest"),  
  deep_learning = c("BERT", "RoBERTa", "LLaMA"),  
  topic_models  = c("LDA", "STM", "BERTopic"),  
  classical_nlp = c("TF-IDF", "n-grams", "Named Entity Recognition")  
)
```

Method Family	Examples
Lexicon-based	AFINN, Bing, NRC, VADER
ML (Supervised)	Naive Bayes, SVM, Random Forest
Deep Learning / LLM	BERT, RoBERTa, GPT-4
Topic Models	LDA, STM, BERTopic
Classical NLP	TF-IDF, n-grams, NER

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Lexicon-Based Sentiment Analysis in R

tidytext pipeline with multiple lexicons:

```
library(tidytext)
library(dplyr)

# AFINN: integer scores -5 to +5
afinn_scores <- tokens_df |>
  inner_join(get_sentiments("afinn"), by = "word") |>
  group_by(post_id) |>
  summarise(sentiment_score = sum(value))

# Bing: binary positive/negative
bing_counts <- tokens_df |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(post_id, sentiment) |>
  pivot_wider(names_from = sentiment,
              values_from = n, values_fill = 0) |>
  mutate(net = positive - negative)
```

Topic Modeling: Latent Dirichlet Allocation (LDA)

Core intuition: Documents are mixtures of topics; topics are distributions over words.

```
library(topicmodels)
library(tidytext)

# Create document-term matrix
dtm <- tokens_df |>
  count(post_id, word) |>
  cast_dtm(post_id, word, n)

# Fit LDA with k topics
lda_model <- LDA(dtm, k = 5,
  control = list(seed = 580))

# Extract top terms per topic
lda_model |>
  tidy(matrix = "beta") |>
```

TF-IDF: Identifying Distinctive Vocabulary

Term Frequency–Inverse Document Frequency identifies words that are important to a document *relative to a corpus*:

$$\text{TF-IDF}(t, d) = \underbrace{\frac{f_{t,d}}{\sum_k f_{k,d}}}_{\text{TF}} \times \underbrace{\log \frac{N}{|\{d : t \in d\}|}}_{\text{IDF}}$$

```
brand_tfidf <- tokens_df |>
  count(brand, word) |>
  bind_tf_idf(word, brand, n) |>
  group_by(brand) |>
  slice_max(tf_idf, n = 10) |>
  ggplot(aes(tf_idf,
              reorder_within(word, tf_idf, brand))) +
  geom_col(fill = "#003D79") +
  facet_wrap(~brand, scales = "free")
```


Section 3

Part III: Potential — What Automated Methods Do Well

Scale and Speed: The Core Advantage

By the Numbers

A trained human coder can analyze ~**200–400** social posts per hour.
An R pipeline can analyze **200,000+** posts per minute.

Real-world applications where scale matters:

- Brand tracking across millions of mentions
- Crisis detection (sudden sentiment shift)
- Competitive intelligence monitoring
- Policy event impact analysis (e.g., FDA announcement → Reddit posts)
- Election forecasting via search & social volume

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Automated methods offer key scientific advantages:

Human Coding

- Inter-rater disagreement
- Coder fatigue effects
- Hard to fully replicate
- Training drift over time

Automated Methods

- 100% consistent rules
- Codebook = the code itself
- Fully reproducible
- Versioned, auditable

Caveat

Reproducibility $\neq$ validity. A machine can be consistently *wrong*.

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Longitudinal Tracking and Event Studies

Combining sentiment with event timelines:

```
library(ggplot2)
library(lubridate)

# Plot daily sentiment around product launch
sentiment_daily |>
  ggplot(aes(x = date, y = mean_sentiment)) +
  geom_line(color = "#003D79", linewidth = 1) +
  geom_ribbon(aes(ymin = ci_low, ymax = ci_high),
            alpha = 0.2, fill = "#003D79") +
  geom_vline(xintercept = as.Date("2024-09-15"),
            linetype = "dashed",
            color = "#DC5032") +
  annotate("text", x = as.Date("2024-09-15"),
            y = max(sentiment_daily$mean_sentiment),
            label = "Product Launch",
            hjust = -0.1, color = "#DC5032") +
```


Section 4

Part IV: Limitations — Where Automated Methods Struggle

The Validity Problem

Are we measuring what we think we're measuring?

Consider these posts - all about the same brand:

```
posts <- c(  
  "This phone is sick!",           # Slang = positive  
  "Not bad for the price",         # Negation + sentiment  
  "I can't believe how good this is", # Double negation  
  "Sure, if you like paying $1000 for a brick", # Sarcasm  
  "My grandmother loves it "      # Emoji irony  
)
```

AFINN/Bing will misclassify most of these

Common Failure Modes

- Slang & colloquialisms
- Sarcasm & irony
- Negation handling
- Domain-specific language

Emoji & Multimodal

- Emoji sentiment reversal
- Memes with no text
- Code-switching languages
- Image-based humor

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Lexicon Limitations: Context Blindness

AFINN assigns fixed scores regardless of context:

Phrase	AFINN Score	Actual Sentiment
"This diet is sick"	-3	Positive (slang)
"Not bad"	-3	Mildly positive
"Killing it in sales"	-3	Very positive
"Surprisingly good"	+3	Positive
"Could be worse"	-2	Mildly positive

Key Limitation

Bag-of-words approaches discard word order, context, and pragmatics. Each word is treated as independent — a flawed assumption for natural language.

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Sampling and Representation Bias

Who is actually posting on social media?

- **Platform demographics** differ drastically from the general population
- **Vocal minority effect:** Most users lurk; a few post frequently
- **Bot contamination:** Estimated 5–15% of Twitter/X accounts are automated
- **Survivorship bias:** Deleted and private posts are invisible
- **Amplification algorithms:** What gets scraped $\neq$ what was written

Methodological Implication

Social media samples are **not random samples** of consumer opinion. Findings may not generalize to the broader market.
(Jungherr et al., 2017; Hargittai, 2020)

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The API Access Problem

Researcher access to social media data has eroded significantly:

- **Twitter/X (2023):** Free Academic API discontinued; basic tier: 500K tweets/month
- **Reddit (2023):** Third-party API costs up 10,000%; killed pushshift
- **TikTok:** No public research API; scrapers frequently broken
- **Meta:** CrowdTangle shut down (August 2024)

Alternative Platforms

- **Bluesky:** Open AT Protocol (atrrr pkg)
- **Mastodon:** Open API
- **YouTube:** Still accessible via tuber
- **Reddit:** Limited free tier remains

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Four key ethical dimensions (franzke et al., 2020):

1 Consent & Expectation of Privacy

Public $\neq$ consented to research use. Context matters (Twitter vs. support groups).

2 Potential for Harm

Exposing vulnerable users, doxxing through data linkage, re-identification.

3 Intellectual Property

Platform ToS, copyright of user-generated content.

4 Algorithmic Bias

NLP models trained on biased corpora reproduce and amplify those biases.

IRB review is required even for "public" social media data in most institutions.
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Section 5

Part V: Best Practices and Validation

The Gold Standard: Human + Machine

Recommended hybrid validation workflow:

```
# Step 1: Draw a stratified random sample
set.seed(580)
validation_sample <- posts_df |>
  slice_sample(n = 200, weight_by = abs(sentiment_score))

# Step 2: Export for human coding
write_csv(validation_sample, "for_human_coding.csv")

# Step 3: Compare machine vs. human labels
validation_results <- validation_sample |>
  left_join(human_codes, by = "post_id") |>
  mutate(agreement = machine_label == human_label)

# Step 4: Report agreement statistics
caret::confusionMatrix(
  factor(validation_results$machine_label)
```

Choosing the Right Method for Your Research Question

If you need...	Use...	Avoid...
Quick sentiment at scale	AFINN / Bing	LLM APIs (cost)
Nuanced emotion	NRC / VADER	Simple polarity
Topic discovery	LDA / BERTopic	Keyword lists
Domain-specific terms	Train custom lexicon	Off-shelf lexicons
High accuracy on small N	Fine-tuned BERT	Bag-of-words
Exploratory/unknown text	TF-IDF + topic model	Supervised ML

Practical Rule

Start simple. Validate against human judgment. Escalate complexity only when simpler methods fail on your domain.

Minimum reporting requirements (following Grimmer & Stewart, 2013):

- **Data provenance:** Platform, date range, API version, filters applied
- **Preprocessing decisions:** Stopwords, stemming/lemmatization, minimum frequency
- **Method justification:** Why this lexicon/model for this domain?
- **Validation evidence:** Agreement statistics (Cohen's κ , F1, accuracy) on held-out human-coded sample
- **Limitations disclosed:** Platform representativeness, known failure modes
- **Code availability:** Reproducible pipeline (GitHub preferred)

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Section 6

Part VI: Emerging Frontiers

Large Language Models as Content Analysts

LLMs (GPT-4, Claude, LLaMA) offer new capabilities:

Zero-shot classification via Anthropic API

`library(httr2)`

```
classify_sentiment <- function(text) {  
  req <- request("https://api.anthropic.com/v1/messages") |>  
    req_headers(  
      "x-api-key" = Sys.getenv("ANTHROPIC_API_KEY"),  
      "anthropic-version" = "2023-06-01"  
    ) |>  
    req_body_json(list(  
      model = "claude-sonnet-4-6",  
      max_tokens = 10,  
      messages = list(list(  
        role = "user",  
        content = paste0("Classify sentiment as positive,",  
                          " negative, or neutral. Reply with one word only.")  
      )  
    )  
  )  
}
```


LLMs: Promise and Peril for Social Media Research

Advantages

- Handle sarcasm, irony, context
- Multilingual out-of-box
- Zero-shot for new tasks
- Nuanced emotion detection
- Structured output extraction

Limitations

- Cost at scale (API pricing)
- Stochastic — not reproducible
- Model version drift
- Hallucination risk
- Black-box interpretability
- Data privacy concerns

Research Frontier

Can we use LLMs to *validate* traditional methods rather than replace them? (Pangakis et al., 2023)

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BERTopic and Neural Topic Models

Moving beyond LDA: embedding-based topic discovery

Key advantages of neural topic models:

- 1 **Semantic coherence:** Topics based on meaning, not co-occurrence
- 2 **Dynamic topics:** Track how topics evolve over time
- 3 **Outlier handling:** Documents don't need to fit a topic
- 4 **No hyperparameter search:** HDBSCAN finds k automatically

```
# BERTopic via reticulate in R
library(reticulate)
bt <- import("bertopic")
model <- bt$BERTopic(language = "english",
                      calculate_probabilities = TRUE)
topics <- model$fit_transform(docs)
model$get_topic_info()
```


Section 7

Part VII: Summary and Discussion

Key Takeaways

What we covered today:

- 1 Social media text is a rich but **non-representative** data source
- 2 Lexicon-based methods are **fast and transparent**, but context-blind
- 3 Topic models reveal **latent structure** without supervision
- 4 LLMs offer **higher validity** but raise cost, reproducibility, and privacy concerns
- 5 **Validation against human judgment** is non-negotiable
- 6 **API access is fragile** — open platforms (Bluesky, YouTube) are your friends
- 7 **Ethics** must be built into the research design, not added at the end

Xu's Rule

"Your AI is only as good as the data it is built on."

Discussion Questions

For reflection and class discussion:

- ➊ A brand manager asks you to measure consumer sentiment using Instagram comments. What are the three biggest threats to validity in this study?
- ➋ When would you prefer AFINN over a fine-tuned BERT model for a marketing research project? When would you switch?
- ➌ A competitor appears to be flooding Reddit with positive posts about their product. How would automated methods detect — or miss — this?
- ➍ How should researchers disclose the use of LLMs (e.g., GPT-4) as part of a content analysis pipeline in a peer-reviewed paper?

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